

FINAL

ST BENEDICT'S

SUBJECT

Advanced Programme
Mathematics Paper 2

DATE

9 November 2016

GRADE

11

MARKS

100

EXAMINER

Mrs MH Povall

MODERATORS

Mrs Sillman, Mr Benecke, Mrs
Serafino

NAME

MEMO

DURATION

1.5 hour

TEACHER

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1	Line of regression	13	
2	Probability Trees	8	
3	Conditional probability	12	
4	Normal distribution	13	
5-6	Binomial and hypergeometric distributions	26	
7	Combinations and permutations	28	
TOTAL		100	

INSTRUCTIONS:

1. Write your name and your Mathematics teacher's name on this test.
2. Answer all questions in the answer booklet provided.
3. Show all working out, as answers only will not guarantee you full marks.

QUESTION 1

13 MARKS

A Ballito beach vendor kept the records of the number of ice-creams sold and the midday temperatures over a 10 day period. Unfortunately something was spilt over his results and some data was lost.

Temperature (°C), x	25	26	16	24	18	21	18	19	28	23
Ice-cream sales, y	305	373	77	316						

The following information had been calculated before the data was lost:

$$\bar{x} = 21.8; \bar{y} = 228.3; \sum x = 218; \sum y = 2283; \sum x^2 = 4896; \sum y^2 = 634819; \sum xy = 53563$$

and $r = 0.939$

- a) What did the correlation coefficient tell us about the relationship between temperature and ice-cream sales.

Very strong positive linear correlation

(3)

(3)

- b) Determine the equation of the least squares regression line for this data. You may use the fact that this line passes through the point $(\bar{x}; \bar{y})$

(4)

$$\begin{aligned}
 b &= \frac{n \sum(xy) - \sum x \sum y}{n(\sum x^2) - (\sum x)^2} \\
 &= \frac{10(53563) - (218)(2283)}{10(4896) - (218)^2} \\
 &= 26,42 \checkmark \checkmark
 \end{aligned}$$

~~any~~

$$y - 228,3 = 26,42(x - 21,8)$$

$$y = 26,42x - 347,66$$

(4)



- c) Estimate the ice-cream sales on days with the following midday temperatures. Comment on your answer in each case in terms of the reliability as an estimate.

i) 20°C

(3)

$$y = 26,42(20) - 347,66 \\ = 180,74 \checkmark$$

Reliable : Interpolation
Correlation coefficient is strong

ii) 38°C

(3)

$$y = 26,42(38) - 347,66 \\ y = 656,30 \checkmark$$

Unreliable : Extrapolated
Correlation coefficient is strong

QUESTION 2

8 MARKS

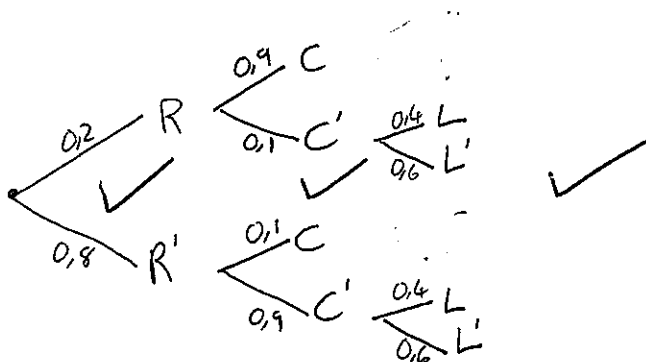
Peter needs to be home by 17:30 on Wednesday but he is due to play a cricket match.

The weather report says there is a 20% chance of rain.

If it does rain, the probability that the match will be cancelled is 90%.

If it is fine, there is a 90% chance that the match will go ahead. The cricket match has a 60% chance of finishing before 17:30, if it is played.

Draw a tree diagram and use it, or otherwise, to calculate the probability that Peter will be home late (after 17:30).



$$P(L) = (0,2 \times 0,1 \times 0,4) + (0,8 \times 0,9 \times 0,4) \\ = 0,296 \checkmark$$

QUESTION 3



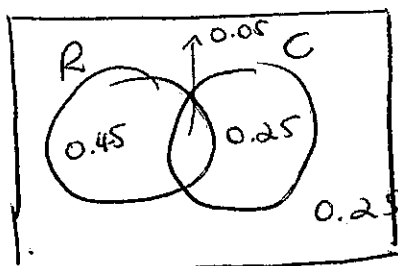
12 MARKS

A survey was undertaken at St Benedict's to determine the proportion of learners who play cricket and rugby. A random sample showed the following:

- The probability that a learner plays rugby is 0,5
- The probability that a learner plays cricket is 0,3
- The probability that a learner plays cricket given that he plays rugby is 0,1.



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- a) Find the probability that a learner plays both sports.

(4)

$$P(C|R) = \frac{P(C \cap R)}{P(R)}$$

$$0,1 = \frac{P(C \cap R)}{0,5}$$

$$P(C \cap R) = 0,05$$

(4)

- b) Calculate the probability that a learner plays neither of these two sports.

(4)

$$P(R \cup C) = P(R) + P(C) - P(R \cap C)$$

$$P(R \cup C) = 0,5 + 0,3 - 0,05$$

$$P(R \cup C) = 0,75$$

$$P(R \cup C)' = 0,25$$

or 2 marks for Venn
+ 2 Ans.

(4)

- c) Are the two events independent? Support your answer using calculations.

(4)

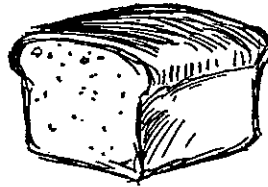
$$P(R \cap C) = 0,05$$

$$P(R) \times P(C) = 0,5 \times 0,3 = 0,15$$

$$P(R \cap C) \neq P(R) \times P(C)$$

Not independent

(4)



QUESTION 4

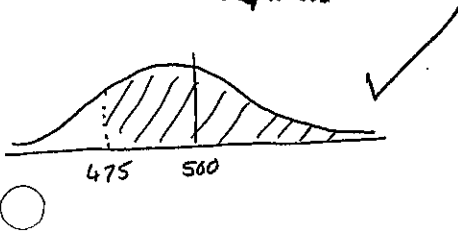
13MARKS

The mass of a small loaf of bread is normally distributed with mean 500g and standard deviation 20g.

Find the probability that a randomly chosen loaf of bread has a mass :

(Round your answer to 4 decimal places)

- a) ^{greater}
not less than 475g

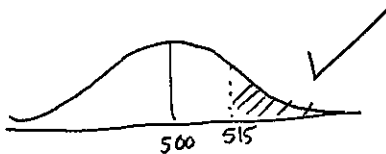


$$\begin{aligned} P(X > 475) \\ &= P(Z > -1,25) \\ &= 0,3944 + 0,5 \\ &= 0,8944 \end{aligned}$$

$$\begin{aligned} Z &= \frac{475 - 500}{20} \\ &= -1,25 \end{aligned}$$

(4)

- b) at least 515g

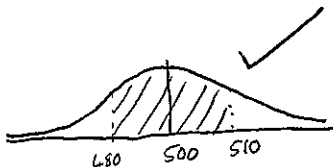


$$\begin{aligned} P(X > 515) \\ &= P(Z > 0,75) \\ &= 0,5 - 0,2734 \\ &= 0,2266 \end{aligned}$$

$$\begin{aligned} Z &= \frac{515 - 500}{20} \\ &= 0,75 \end{aligned}$$

(4)

- c) between 480g and 510g



$$\begin{aligned} P(480 < X < 510) \\ &= P(-1 < Z < 0,5) \\ &= 0,3413 + 0,1915 \\ &= 0,5328 \end{aligned}$$

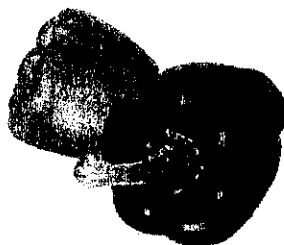
$$\begin{aligned} Z &= \frac{480 - 500}{20} \\ &= -1 \end{aligned}$$

$$\begin{aligned} Z &= \frac{510 - 500}{20} \\ &= 0,5 \end{aligned}$$

(5)

QUESTION 5

15 MARKS



A vegetable basket contains 12 peppers, of which 3 are red, 4 are green and 5 are yellow. Three peppers are taken, at random and without replacement, from the basket.

(Round your answer to 4 decimal places)

- a) Find the probability that the three peppers are all different colours. (5)

$$\frac{\binom{3}{1}\binom{4}{1}\binom{5}{1}}{\binom{12}{3}} = \frac{3}{11} \text{ or } 0,2727 \quad (5)$$

- b) Find the probability of getting the three peppers to be of the same colour. (5)

$$\frac{\binom{3}{3}}{\binom{12}{3}} + \frac{\binom{4}{3}}{\binom{12}{3}} + \frac{\binom{5}{3}}{\binom{12}{3}} = \frac{1}{220} + \frac{4}{220} + \frac{10}{220} = \frac{3}{44} \text{ or } 0,0682 \quad (5)$$

- c) What is the probability that exactly 2 of the peppers taken are green? (5)

$$\frac{\binom{4}{2}\binom{8}{1}}{\binom{12}{3}} = \frac{12}{55} \text{ or } 0,2182 \quad (5)$$

QUESTION 6

11 MARKS

Ballpoint pens are made in large quantities in a factory. They are tested in batches of 50 and it is known that 3% are faulty.

(Round your answer to 4 decimal places)



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(a) From a test batch calculate :

(1) the probability that exactly 2 pens are faulty

(4)

$$\binom{50}{2} (0,03)^2 (0,97)^{48} = 0,2555$$

4

(2) the probability that at least one of the pens are faulty

(5)

$$1 - \binom{50}{0} (0,03)^0 (0,97)^{50} = 0,7819$$

5

(b) In 200 batches of 50, how many pens would you expect to be faulty?

(2)

$$200 \times 50 \times 0,03 = 300$$

2

7
QUESTION 8

28 MARKS

a) The six digits 4, 5, 6, 7, 8, 9 can be arranged to give many 6-digit numbers.

i) How many 6-digit numbers can be made? (2)

$$6! = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{1} = 720 \quad \checkmark \quad (2)$$

ii) How many of these 6-digit numbers start with an odd digit and end with an odd digit? (4)

$$3 \times 2 \times 4! = 144 \quad \checkmark \quad (4)$$

b) Find the number of ways of choosing a school team of 5 pupils from 6 boys and 8 girls

i) if there are 2 boys and 3 girls in the team. (3)

$$\binom{6}{2} \times \binom{8}{3} = 840 \quad \checkmark \quad (3)$$

ii) if there are more girls than boys in the team, (both sexes are represented) (4)

$$3 \text{ girls} : \binom{6}{2} \times \binom{8}{3} = 840 \quad \checkmark$$

$$4 \text{ girls} : \binom{6}{1} \times \binom{8}{4} = 420 \quad \checkmark$$

$$5 \text{ girls} : \binom{6}{0} \times \binom{8}{5} = 56 \quad \checkmark$$

$$840 + 420 + 56 = 1316 \quad \checkmark \quad (4)$$

- c) Eight girls have won prizes in a bikini contest on the beach. They may each select a prize, and their choice is limited to a beach ball, a t-shirt or a beach umbrella. There are sufficient items for them all to make the same choice.



- i) How many different prize selections are possible? (2)

$$3^8 = 6561 \quad \checkmark \quad \textcircled{2}$$

- ii) How many ways can ^{the word} BIKINI be arranged? (3)

$$\frac{6!}{3!} = 120 \quad \checkmark \quad \textcircled{3}$$

- iii) What is the probability of the new word formed starting and ending with the same letter? (4) (5)

$$\frac{4!}{120} = \frac{24}{120} = \frac{1}{5} \text{ or } 0,2 \quad \checkmark \quad \textcircled{4}$$

- iv) What is the probability that the letters K and N will be next to each other? (5)

$$\frac{2! \times 5!}{3!} = \frac{40}{120} = \frac{1}{3} \text{ or } 0,33 \quad \checkmark \quad \textcircled{5}$$

